



## EXACT DYNAMIC STIFFNESS MATRIX FOR COMPOSITE TIMOSHENKO BEAMS WITH APPLICATIONS

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In this paper, an exact dynamic stiffness matrix is presented for a composite beam. It includes the effects of shear deformation and rotatory inertia: i.e., it is for a composite Timoshenko beam. The theory accounts for the (material) coupling between the bending and torsional deformations which usually occurs for such beams due to the anisotropic nature of fibrous composites. An explicit analytical expression for each of the elements of the dynamic stiffness matrix is derived by rigorous use of the symbolic computing package *REDUCE*. It is proved that the use of such expressions leads to substantial savings in computer time when compared with the matrix inversion method. The use of this dynamic stiffness matrix to investigate the free vibration characteristics of composite beams (with or without the effects of shear deformation and/or rotatory inertia included) is demonstrated by applying the Wittrick–Williams algorithm. Numerical results for which comparative results are available in the literature are discussed.

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### 1. INTRODUCTION

The free bending or bending–torsion coupled vibration characteristics of composite beams have been investigated by a number of authors [1–16], of whom only a few [2, 4, 9, 13, 14] have focused attention on the effects of shear deformation and rotatory inertia: i.e., on composite Timoshenko beams. Chandrashekhara *et al.* [9], Abramovich [13] and Abramovich and Livshits [14] considered free bending vibration of composite Timoshenko beams, but without including the (material) coupling between the bending and torsional deformations. In contrast, Teoh and Huang [2] and Teh and Huang [4] included this coupling in their investigation.

The literature on free vibration analysis of composite beams is concentrated primarily on single straight uniform composite beams with some prescribed boundary (end) conditions, rather than on assemblies of such beams. In contrast, the authors recently obtained the dynamic stiffness matrix for a uniform, straight, bending–torsion (materially) coupled, composite beam [16]. Hence they gave a general method which can be used, for example, to model a non-uniform or tapered composite beam as an assembly of many uniform composite beams; i.e., as a stepped composite beam. Of course, an incidental benefit is that the free vibration behaviour of a single individual uniform composite beam

with any desired boundary conditions can be studied by using the appropriate stiffness terms of this dynamic stiffness matrix; e.g., by deleting those rows and columns of the matrix corresponding to zero boundary displacements.

However, the recently developed dynamic stiffness matrix of Banerjee and Williams [16] did not account for the effects of shear deformation and rotatory inertia, which can be important for composite beams [6] because they are usually much more sensitive to these effects than are their metallic counterparts, due mainly to the low shear moduli of fibrous composites. The task of extending this dynamic stiffness matrix of a bending–torsion coupled composite beam to include the effects of shear deformation and rotatory inertia, i.e., to cover composite Timoshenko beams, is reported in the present paper. It will be seen that considerable analytical and computational efforts were required, particularly when deriving explicit analytical expressions for the elements of the dynamic stiffness matrix. Extensive use has been made of the recent advances in symbolic computing. Comparative results obtained from the present theory for free vibration characteristics of an appropriately chosen composite beam from the literature are presented and discussed.

## 2. THEORY

A straight uniform composite beam with length  $L$ , solid rectangular cross-section and symmetric but unbalanced lay-up is shown in Figure 1. Bending–torsion coupling is well known to occur for such configurations [2–5, 7, 8, 16]. However, the theory which follows applies to more general cross-sections as discussed in section 3.2, but the rectangular cross-section is shown for convenience. In the right-hand co-ordinate system of Figure 1, the elastic axis, which is assumed to coincide with the  $y$ -axis, is permitted flexural translation  $h(y, t)$  in the  $z$ -direction and torsional rotation  $\psi(y, t)$  about the  $y$ -axis, where  $y$  and  $t$  denote distance from the origin and time, respectively.

The governing partial differential equations of motion for the coupled bending–torsional free natural vibration of the composite beam of Figure 1 with shear deformation and rotatory inertia included, but with warping stiffness neglected, are given by [2, 4]

$$EI\theta'' + kAG(h' - \theta) + K\psi'' - \rho I\ddot{\theta} = 0, \quad (1)$$

$$kAG(h'' - \theta') - m\ddot{h} = 0, \quad GJ\psi'' + K\theta'' - I_z\ddot{\psi} = 0, \quad (2, 3)$$

where primes and dots denote differentiation with respect to position  $y$  and time  $t$  respectively;  $\rho$  is the density of the material;  $EI$ ,  $GJ$ ,  $K$  and  $kAG$  are, respectively, the

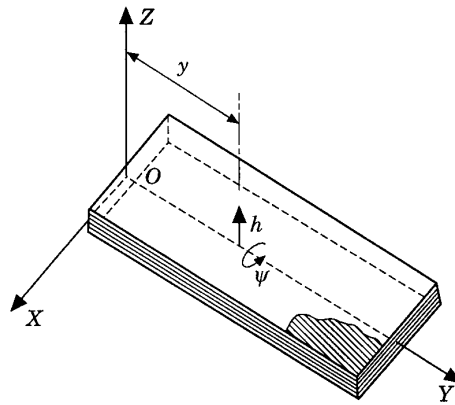


Figure 1. The co-ordinate system and notation for a bending–torsion coupled composite Timoshenko beam.

bending rigidity, torsional rigidity, bending–torsion coupling rigidity and shear rigidity of the composite beam;  $I$  is the second moment of area of the beam cross-section about the  $x$ -axis;  $m$  is the mass per unit length,  $=\rho A$ , where  $A$  is the cross-sectional area;  $I_x$  is the polar mass moment of inertia per unit length about the  $y$ -axis; and  $\theta$  is the angle of rotation in radians of the cross-section about the  $x$ -axis due to bending alone, so that the total slope  $h'$  equals the sum of the slopes due to bending and due to shear deformation. Note that  $EI$ ,  $GJ$ ,  $K$  and  $kAG$ , which characterize a bending–torsion (materially) coupled composite beam, can be found either theoretically [11, 17] or experimentally [18].

If a sinusoidal variation of  $h$ ,  $\theta$  and  $\psi$ , with circular frequency  $\omega$ , is assumed, then

$$h(y, t) = H(y) \sin \omega t, \quad \theta(y, t) = \Theta(y) \sin \omega t, \quad \psi(y, t) = \Psi(y) \sin \omega t, \quad (4)$$

where  $H(y)$ ,  $\Theta(y)$  and  $\Psi(y)$  are the amplitudes of the sinusoidally varying vertical displacement, bending rotation and twist respectively.

Substituting equation (4) into equations (1)–(3) gives

$$EI\Theta'' + kAG(H' - \Theta) + K\Psi'' + \rho I\omega^2\Theta = 0, \quad (5)$$

$$kAG(H'' - \Theta') + m\omega^2H = 0, \quad GJ\Psi'' + K\Theta'' + I_x\omega^2\Psi = 0. \quad (6, 7)$$

By extensive algebraic manipulation, equations (5)–(7) can be combined into one equation, by eliminating all but one of the three variables  $H$ ,  $\Theta$  and  $\Psi$ , to give

$$(D^6 + \bar{a}D^4 - \bar{b}D^2 - \bar{c})W = 0, \quad (8)$$

where

$$W = H, \Theta \text{ or } \Psi \quad (9)$$

$$D = d/d\xi, \quad \xi = y/L, \quad (10)$$

and

$$\begin{aligned} \bar{a} &= \{a^2 + b^2(r^2 + s^2c^2)\}/c^2, \\ \bar{b} &= b^2\{(1 - b^2r^2s^2) - a^2(r^2 + s^2)\}/c^2, \\ \bar{c} &= a^2b^2(1 - b^2r^2s^2)/c^2, \end{aligned} \quad (11)$$

with

$$\begin{aligned} a^2 &= I_x\omega^2L^2/GJ, \quad b^2 = m\omega^2L^4/EI, \quad c^2 = 1 - K^2/EIGJ, \\ r^2 &= I/AL^2, \quad s^2 = EI/kAGL^2. \end{aligned} \quad (12)$$

The solution of the differential equation (8) is [2, 4, 16]

$$W(\xi) = C_1^* \cosh \alpha\xi + C_2^* \sinh \alpha\xi + C_3^* \cos \beta\xi + C_4^* \sin \beta\xi + C_5^* \cos \gamma\xi + C_6^* \sin \gamma\xi, \quad (13)$$

where  $C_1^*$ – $C_6^*$  are constants and

$$\begin{aligned} \alpha &= [2(q/3)^{1/2} \cos(\phi/3) - \bar{a}/3]^{1/2}, \quad \beta = [2(q/3)^{1/2} \cos\{(\pi - \phi)/3\} + \bar{a}/3]^{1/2}, \\ \gamma &= [2(q/3)^{1/2} \cos\{(\pi + \phi)/3\} + \bar{a}/3]^{1/2}, \end{aligned} \quad (14)$$

with

$$q = \bar{b} + \bar{a}^2/3, \quad \phi = \cos^{-1} [(27\bar{c} - 9\bar{a}\bar{b} - 2\bar{a}^3)/\{2(\bar{a}^2 + 3\bar{b})^{3/2}\}]. \quad (15)$$

Equation (13) represents the solution for the bending displacement  $H(\xi)$ , bending rotation  $\Theta(\xi)$  and torsional rotation  $\Psi(\xi)$ . Thus

$$H(\xi) = A_1 \cosh \alpha\xi + A_2 \sinh \alpha\xi + A_3 \cos \beta\xi + A_4 \sin \beta\xi + A_5 \cos \gamma\xi + A_6 \sin \gamma\xi, \quad (16)$$

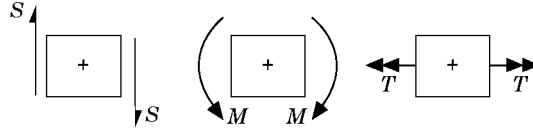


Figure 2. The sign convention for positive transverse (shear) force ( $S$ ), bending moment ( $M$ ) and torque ( $T$ ).

$$\Theta(\xi) = B_1 \sinh \alpha \xi + B_2 \cosh \alpha \xi + B_3 \sin \beta \xi + B_4 \cos \beta \xi + B_5 \sin \gamma \xi + B_6 \cos \gamma \xi, \quad (17)$$

$$\Psi(\xi) = C_1 \cosh \alpha \xi + C_2 \sinh \alpha \xi + C_3 \cos \beta \xi + C_4 \sin \beta \xi + C_5 \cos \gamma \xi + C_6 \sin \gamma \xi, \quad (18)$$

where  $A_1$ – $A_6$ ,  $B_1$ – $B_6$  and  $C_1$ – $C_6$  are three different sets of constants.

Substituting equations (16) and (17) into equation (6) shows that

$$\begin{aligned} B_1 &= (\bar{\alpha}/L)A_1, & B_3 &= -(\bar{\beta}/L)A_3, & B_5 &= -(\bar{\gamma}/L)A_5, \\ B_2 &= (\bar{\alpha}/L)A_2, & B_4 &= (\bar{\beta}/L)A_4, & B_6 &= (\bar{\gamma}/L)A_6, \end{aligned} \quad (19)$$

where

$$\bar{\alpha} = (\alpha^2 + b^2 s^2)/\alpha, \quad \bar{\beta} = (\beta^2 - b^2 s^2)/\beta, \quad \bar{\gamma} = (\gamma^2 - b^2 s^2)/\gamma. \quad (20)$$

Then substituting equations (17) and (18) into equation (7) and using equations (19) gives

$$\begin{aligned} C_1 &= (k_\alpha/L)A_2, & C_3 &= (k_\beta/L)A_4, & C_5 &= (k_\gamma/L)A_6, \\ C_2 &= (k_\alpha/L)A_1, & C_4 &= -(k_\beta/L)A_3, & C_6 &= -(k_\gamma/L)A_5, \end{aligned} \quad (21)$$

where

$$\begin{aligned} k_\alpha &= -k_t \{ \alpha(\alpha^2 + b^2 s^2)/(\alpha^2 + a^2) \}, \\ k_\beta &= -k_t \{ \beta(\beta^2 - b^2 s^2)/(\beta^2 - a^2) \}, & k_\gamma &= -k_t \{ \gamma(\gamma^2 - b^2 s^2)/(\gamma^2 - a^2) \}, \end{aligned} \quad (22)$$

with

$$k_t = K/GJ. \quad (23)$$

By following the sign convention of Figure 2, the expressions for the bending moment  $M(\xi)$ , shear force  $S(\xi)$  and torque  $T(\xi)$  are obtained from equations (16)–(18), after some simplification, as

$$\begin{aligned} M(\xi) &= -(EI/L) (d\Theta/d\xi) - (K/L)(d\Psi/d\xi) = -(EI/L^2) \{ A_1 g_\alpha \cosh \alpha \xi + A_2 g_\alpha \sinh \alpha \xi \\ &\quad - A_3 g_\beta \cos \beta \xi - A_4 g_\beta \sin \beta \xi - A_5 g_\gamma \cos \gamma \xi - A_6 g_\gamma \sin \gamma \xi \}, \end{aligned} \quad (24)$$

$$\begin{aligned} S(\xi) &= (EI/L^2) [d^2\Theta/d\xi^2 + (K/EI) (d^2\Psi/d\xi^2) + b^2 r^2 \Theta] \\ &= (EI/L^3) \{ A_1 f_\alpha \sinh \alpha \xi + A_2 f_\alpha \cosh \alpha \xi + A_3 f_\beta \sin \beta \xi - A_4 f_\beta \cos \beta \xi \\ &\quad + A_5 f_\gamma \sin \gamma \xi - A_6 f_\gamma \cos \gamma \xi \}, \end{aligned} \quad (25)$$

$$\begin{aligned} T(\xi) &= (GJ/L) [(d\Psi/d\xi) + k_t (d\Theta/d\xi)] \\ &= (GJ/L^2) \{ A_1 e_\alpha \cosh \alpha \xi + A_2 e_\alpha \sinh \alpha \xi \\ &\quad - A_3 e_\beta \cos \beta \xi - A_4 e_\beta \sin \beta \xi - A_5 e_\gamma \cos \gamma \xi - A_6 e_\gamma \sin \gamma \xi \}, \end{aligned} \quad (26)$$

where

$$g_\alpha = \alpha(\bar{\alpha} + k_b k_\alpha), \quad g_\beta = \beta(\bar{\beta} + k_b k_\beta), \quad g_\gamma = \gamma(\bar{\gamma} + k_b k_\gamma), \quad (27)$$

$$f_\alpha = \alpha g_\alpha \bar{\alpha} b^2 r^2, \quad f_\beta = \beta g_\beta - \bar{\beta} b^2 r^2, \quad f_\gamma = \gamma g_\gamma - \bar{\gamma} b^2 r^2, \quad (28)$$

$$e_\alpha = \alpha(k_\alpha + \bar{\alpha}k_t), \quad e_\beta = \beta(k_\beta + \bar{\beta}k_t), \quad e_\gamma = \gamma(k_\gamma + \bar{\gamma}k_t), \quad (29)$$

with

$$k_b = K/EI. \quad (30)$$

The end conditions for displacements and forces of the beam element (see Figure 3) are, respectively,

$$\begin{aligned} \text{at end 1 (i.e. } \xi = 0); H = H_1, \quad \Theta = \Theta_1 \quad \text{and} \quad \Psi = \Psi_1; \\ \text{at end 2 (i.e. } \xi = 1); H = H_2, \quad \Theta = \Theta_2 \quad \text{and} \quad \Psi = \Psi_2; \end{aligned} \quad (31)$$

$$\begin{aligned} \text{at end 1 (i.e. } \xi = 0); S = S_1, \quad M = M_1 \quad \text{and} \quad T = -T_1; \\ \text{at end 2 (i.e. } \xi = 1); S = -S_2, \quad M = -M_2 \quad \text{and} \quad T = T_2. \end{aligned} \quad (32)$$

The dynamic stiffness matrix which relates the amplitudes of the sinusoidally varying forces to the corresponding displacement amplitudes can now be derived with the help of equations (16)–(18), (24)–(26), (31) and (32) as follows.

Substituting equations (31), (19) and (21) in equations (16)–(18) gives

$$\begin{bmatrix} H_1 \\ \Theta_1 \\ \Psi_1 \\ H_2 \\ \Theta_2 \\ \Psi_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & \bar{\alpha}/L & 0 & \bar{\beta}/L & 0 & \bar{\gamma}/L \\ 0 & k_\alpha/L & 0 & k_\beta/L & 0 & k_\gamma/L \\ C_{hx} & S_{hx} & C_\beta & S_\beta & C_\gamma & S_\gamma \\ \bar{\alpha}S_{hx}/L & \bar{\alpha}C_{hx}/L & -\bar{\beta}S_\beta/L & \bar{\beta}C_\beta/L & -\bar{\gamma}S_\gamma/L & \bar{\gamma}C_\gamma/L \\ k_\alpha S_{hx}/L & k_\alpha C_{hx}/L & -k_\beta S_\beta/L & k_\beta C_\beta/L & -k_\gamma S_\gamma/L & k_\gamma C_\gamma/L \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \\ A_3 \\ A_4 \\ A_5 \\ A_6 \end{bmatrix}. \quad (33)$$

i.e.,

$$\mathbf{U} = \mathbf{BA}, \quad (34)$$

where

$$\begin{aligned} C_{hx} = \cosh \alpha, \quad C_\beta = \cos \beta, \quad C_\gamma = \cos \gamma, \\ S_{hx} = \sinh \alpha, \quad S_\beta = \sin \beta, \quad S_\gamma = \sin \gamma. \end{aligned} \quad (35)$$

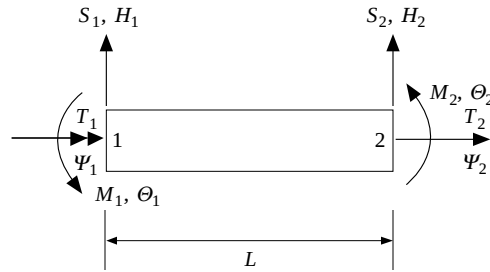


Figure 3. End conditions for forces and displacements of the beam element.

Substituting equations (32) in equations (24)–(26) gives

$$\begin{bmatrix} S_1 \\ M_1 \\ T_1 \\ S_2 \\ M_2 \\ T_2 \end{bmatrix} = \begin{bmatrix} 0 & W_3 f_\alpha & 0 & -W_3 f_\beta & 0 & -W_3 f_\gamma \\ -W_2 g_\alpha & 0 & W_2 g_\beta & 0 & W_2 g_\gamma & 0 \\ -W_1 e_\alpha/L & 0 & W_1 e_\beta/L & 0 & W_1 e_\gamma/L & 0 \\ -W_3 f_\alpha S_{hz} & -W_3 f_\alpha C_{hz} & -W_3 f_\beta S_\beta & W_3 f_\beta C_\beta & -W_3 f_\gamma S_\gamma & W_3 f_\gamma C_\gamma \\ W_2 g_\alpha C_{hz} & W_2 g_\alpha S_{hz} & -W_2 g_\beta C_\beta & -W_2 g_\beta S_\beta & -W_2 g_\gamma C_\gamma & -W_2 g_\gamma S_\gamma \\ W_1 e_\alpha C_{hz}/L & W_1 e_\alpha S_{hz}/L & -W_1 e_\beta C_\beta/L & -W_1 e_\beta S_\beta/L & -W_1 e_\gamma C_\gamma/L & -W_1 e_\gamma S_\gamma/L \end{bmatrix} \times \begin{bmatrix} A_1 \\ A_2 \\ A_3 \\ A_4 \\ A_5 \\ A_6 \end{bmatrix}; \quad (36)$$

i.e.,

$$\mathbf{F} = \mathbf{DA}, \quad (37)$$

where

$$W_1 = GJ/L, \quad W_2 = EI/L^2, \quad W_3 = EI/L^3. \quad (38)$$

Equations (34) and (37) give

$$\mathbf{F} = \mathbf{KU}: \quad (39)$$

i.e.,

$$\begin{bmatrix} S_1 \\ M_1 \\ T_1 \\ S_2 \\ M_2 \\ T_2 \end{bmatrix} = \begin{bmatrix} K_{1,1} & K_{1,2} & K_{1,3} & K_{1,4} & K_{1,5} & K_{1,6} \\ & K_{2,2} & K_{2,3} & K_{2,4} & K_{2,5} & K_{2,6} \\ & & K_{3,3} & K_{3,4} & K_{3,5} & K_{3,6} \\ & \text{symmetric} & & K_{4,4} & K_{4,5} & K_{4,6} \\ & & & & K_{5,5} & K_{5,6} \\ & & & & & K_{6,6} \end{bmatrix} \begin{bmatrix} H_1 \\ \Theta_1 \\ \Psi_1 \\ H_2 \\ \Theta_2 \\ \Psi_2 \end{bmatrix}, \quad (40)$$

where

$$\mathbf{K} = \mathbf{DB}^{-1} \quad (41)$$

is the required stiffness matrix.

The formidable task of inverting the matrix  $\mathbf{B}$  algebraically and then premultiplying by the  $\mathbf{D}$  matrix has only become possible due to recent developments in symbolic computing [19]. Thus most of the work was done by the symbolic computing package *REDUCE* [20] when generating and, more importantly, when simplifying the explicit expressions for the terms of the dynamic stiffness matrix. Such expressions are particularly useful when some but not all of the stiffness coefficients are needed. The expressions derived are presented below in concise form in equations (70)–(81), with appropriate substitutions needed from equations (42)–(69).

Let the following variables be introduced, in which  $\alpha$ ,  $\beta$  and  $\gamma$ ,  $\bar{\alpha}$ ,  $\bar{\beta}$  and  $\bar{\gamma}$ ,  $k_\alpha$ ,  $k_\beta$  and  $k_\gamma$ ,  $g_\alpha$ ,  $g_\beta$  and  $g_\gamma$ ,  $f_\alpha$ ,  $f_\beta$ ,  $f_\gamma$ , and  $e_\alpha$ ,  $e_\beta$  and  $e_\gamma$  are respectively given by equations (14), (20), (22), (27), (28) and (29):

$$\mu_1 = \bar{\alpha}k_\beta - \bar{\beta}k_\alpha, \quad \mu_2 = \bar{\beta}k_\gamma - \bar{\gamma}k_\beta, \quad \mu_3 = \bar{\gamma}k_\alpha - \bar{\alpha}k_\gamma, \quad (42)$$

$$\epsilon_1 = f_\alpha k_\beta + f_\beta k_\alpha, \quad \epsilon_2 = f_\beta k_\gamma - f_\gamma k_\beta, \quad \epsilon_3 = f_\gamma k_\alpha + f_\alpha k_\gamma, \quad (43)$$

$$\zeta_1 = f_\alpha \bar{\beta} + f_\beta \bar{\alpha}, \quad \zeta_2 = f_\beta \bar{\gamma} - f_\gamma \bar{\beta}, \quad \zeta_3 = f_\gamma \bar{\alpha} + f_\alpha \bar{\gamma}, \quad (44)$$

$$\eta_1 = g_\alpha + g_\beta, \quad \eta_2 = g_\beta - g_\gamma, \quad \eta_3 = g_\gamma + g_\alpha, \quad (45)$$

$$\bar{\eta}_1 = e_\alpha + e_\beta, \quad \bar{\eta}_2 = e_\beta - e_\gamma, \quad \bar{\eta}_3 = e_\gamma + e_\alpha, \quad (46)$$

$$\lambda_1 = \mu_1 \epsilon_2 - \mu_2 \epsilon_1, \quad \lambda_2 = \mu_2 \epsilon_3 + \mu_3 \epsilon_2, \quad \lambda_3 = \mu_3 \epsilon_1 - \mu_1 \epsilon_3, \quad (47)$$

$$\bar{\lambda}_1 = \mu_1 \zeta_2 - \mu_2 \zeta_1, \quad \bar{\lambda}_2 = \mu_2 \zeta_3 + \mu_3 \zeta_2, \quad \bar{\lambda}_3 = \mu_3 \zeta_1 - \mu_1 \zeta_3, \quad (48)$$

$$\tau_1 = f_\alpha \mu_2 - f_\beta \mu_3 - f_\gamma \mu_1, \quad \tau_2 = \mu_1 \epsilon_1 + \mu_2 \epsilon_2 - \mu_3 \epsilon_3, \quad \tau_3 = \mu_1 \zeta_1 + \mu_2 \zeta_2 - \mu_3 \zeta_3, \quad (49)$$

$$\xi_1 = k_\alpha \mu_3 \eta_2 - k_\beta \mu_2 \eta_3, \quad \xi_2 = k_\beta \mu_1 \eta_3 - k_\gamma \mu_3 \eta_1, \quad \xi_3 = k_\gamma \mu_2 \eta_1 + k_\alpha \mu_1 \eta_2, \quad (50)$$

$$\bar{\xi}_1 = k_\alpha \mu_2 \eta_3 + k_\beta \mu_3 \eta_2, \quad \bar{\xi}_2 = k_\beta \mu_3 \eta_1 - k_\gamma \mu_1 \eta_3, \quad \bar{\xi}_3 = k_\gamma \mu_1 \eta_2 - k_\alpha \mu_2 \eta_1, \quad (51)$$

$$\rho_1 = \bar{\alpha} \mu_3 \eta_2 - \bar{\beta} \mu_2 \eta_3, \quad \rho_2 = \bar{\beta} \mu_1 \eta_3 - \bar{\gamma} \mu_3 \eta_1, \quad \rho_3 = \bar{\gamma} \mu_2 \eta_1 + \bar{\alpha} \mu_1 \eta_2, \quad (52)$$

$$\bar{\rho}_1 = \bar{\alpha} \mu_2 \eta_3 + \bar{\beta} \mu_3 \eta_2, \quad \bar{\rho}_2 = \bar{\beta} \mu_3 \eta_1 - \bar{\gamma} \mu_1 \eta_3, \quad \bar{\rho}_3 = \bar{\gamma} \mu_1 \eta_2 - \bar{\alpha} \mu_2 \eta_1, \quad (53)$$

$$\nu_1 = \bar{\alpha} \mu_3 \bar{\eta}_2 - \bar{\beta} \mu_2 \bar{\eta}_3, \quad \nu_2 = \bar{\beta} \mu_1 \bar{\eta}_3 - \bar{\gamma} \mu_3 \bar{\eta}_1, \quad \nu_3 = \bar{\gamma} \mu_2 \bar{\eta}_1 + \bar{\alpha} \mu_1 \bar{\eta}_2, \quad (54)$$

$$\bar{\nu}_1 = \bar{\alpha} \mu_2 \bar{\eta}_3 + \bar{\beta} \mu_3 \bar{\eta}_2, \quad \bar{\nu}_2 = \bar{\beta} \mu_3 \bar{\eta}_1 - \bar{\gamma} \mu_1 \bar{\eta}_3, \quad \bar{\nu}_3 = \bar{\gamma} \mu_1 \bar{\eta}_2 - \bar{\alpha} \mu_2 \bar{\eta}_1, \quad (55)$$

$$\sigma_1 = -k_\alpha \mu_2 \eta_2 - k_\beta \mu_3 \eta_3 + k_\gamma \mu_1 \eta_1, \quad \sigma_2 = \bar{\alpha} \mu_2 \eta_2 + \bar{\beta} \mu_3 \eta_3 - \bar{\gamma} \mu_1 \eta_1, \quad (56)$$

$$\sigma_3 = \bar{\alpha} \mu_2 \bar{\eta}_2 + \bar{\beta} \mu_3 \bar{\eta}_3 - \bar{\gamma} \mu_1 \bar{\eta}_1.$$

By using the above substitutions, the following further variables are introduced:

$$\Phi_1 = -\tau_1(\mu_1 S_\beta C_\gamma S_{hx} - \mu_2 S_\beta S_\gamma C_{hx} + \mu_3 C_\beta S_\gamma S_{hx}), \quad (57)$$

$$\Phi_2 = \tau_2 S_\beta S_\gamma S_{hx} + \lambda_3 S_{hx}(1 - C_\beta C_\gamma) + \lambda_1 S_\beta(1 - C_\gamma C_{hx}) + \lambda_2 S_\gamma(1 - C_\beta C_{hx}), \quad (58)$$

$$\Phi_3 = -\tau_3 S_\beta S_\gamma S_{hx} - \bar{\lambda}_3 S_{hx}(1 - C_\beta C_\gamma) - \bar{\lambda}_1 S_\beta(1 - C_\gamma C_{hx}) - \bar{\lambda}_2 S_\gamma(1 - C_\beta C_{hx}), \quad (59)$$

$$\Phi_4 = \tau_1(\mu_1 S_\beta S_{hx} - \mu_2 S_\beta S_\gamma + \mu_3 S_\gamma S_{hx}), \quad (60)$$

$$\Phi_5 = -\tau_1\{k_\alpha S_{hx}(C_\beta - C_\gamma) - k_\beta S_\beta(C_\gamma - C_{hx}) + k_\gamma S_\gamma(C_\beta - C_{hx})\}, \quad (61)$$

$$\Phi_6 = \tau_1\{\bar{\alpha} S_{hx}(C_\beta - C_\gamma) - \bar{\beta} S_\beta(C_\gamma - C_{hx}) + \bar{\gamma} S_\gamma(C_\beta - C_{hx})\}, \quad (62)$$

$$\Phi_7 = -\xi_1 S_\beta C_\gamma S_{hx} + \xi_2 S_\beta S_\gamma C_{hx} - \xi_3 C_\beta S_\gamma S_{hx} + \sigma_1 C_\beta C_\gamma C_{hx} - \bar{\xi}_3 C_\beta - \bar{\xi}_1 C_\gamma + \bar{\xi}_2 C_{hx}, \quad (63)$$

$$\Phi_8 = \rho_1 S_\beta C_\gamma S_{hx} + \rho_3 C_\beta S_\gamma S_{hx} - \rho_2 S_\beta S_\gamma C_{hx} + \sigma_2 C_\beta C_\gamma C_{hx} + \bar{\rho}_3 C_\beta + \bar{\rho}_1 C_\gamma - \bar{\rho}_2 C_{hx}, \quad (64)$$

$$\Phi_9 = \xi_1 S_\beta S_{hx} - \xi_2 S_\beta S_\gamma + \xi_3 S_\gamma S_{hx} - \bar{\xi}_3 C_\gamma C_{hx} + \bar{\xi}_2 C_\beta C_\gamma - \bar{\xi}_1 C_\beta C_{hx} + \sigma_1, \quad (65)$$

$$\Phi_{10} = -\rho_1 S_\beta S_{hz} - \rho_3 S_\gamma S_{hz} + \rho_2 S_\beta S_\gamma + \bar{\rho}_3 C_\gamma C_{hz} - \bar{\rho}_2 C_\beta C_\gamma + \bar{\rho}_1 C_\beta C_{hz} + \sigma_2, \quad (66)$$

$$\Phi_{11} = v_1 S_\beta C_\gamma S_{hz} + v_3 C_\beta S_\gamma S_{hz} - v_2 S_\beta S_\gamma C_{hz} + \sigma_3 C_\beta C_\gamma C_{hz} + \bar{v}_3 C_\beta + \bar{v}_1 C_\gamma - \bar{v}_2 C_{hz}, \quad (67)$$

$$\Phi_{12} = -v_1 S_\beta S_{hz} - v_3 S_\gamma S_{hz} + v_2 S_\beta S_\gamma + \bar{v}_3 C_\gamma C_{hz} - \bar{v}_2 C_\beta C_\gamma + \bar{v}_1 C_\beta C_{hz} + \sigma_3, \quad (68)$$

$$\Delta = (\mu_1^2 - \mu_2^2 + \mu_3^2) S_\beta S_\gamma S_{hz} - 2\mu_1 \mu_2 S_\beta (1 - C_\gamma C_{hz}) - 2\mu_2 \mu_3 S_\gamma (1 - C_\beta C_{hz}) + 2\mu_1 \mu_3 S_{hz} (1 - C_\beta C_\gamma). \quad (69)$$

Then the terms (elements) of the required dynamic stiffness matrix, see equation (40), are given by

$$K_{1,1} = K_{4,4} = (EI/L^3)(\Phi_1/\Delta), \quad K_{1,2} = -K_{4,5} = (EI/L^2)(\Phi_2/\Delta), \quad (70, 71)$$

$$K_{1,3} = -K_{4,6} = (EI/L^2)(\Phi_3/\Delta), \quad K_{1,4} = (EI/L^3)(\Phi_4/\Delta), \quad (72, 73)$$

$$K_{1,5} = -K_{2,4} = (EI/L^2)(\Phi_5/\Delta), \quad K_{1,6} = -K_{3,4} = (EI/L^2)(\Phi_6/\Delta), \quad (74, 75)$$

$$K_{2,2} = K_{5,5} = (EI/L)(\Phi_7/\Delta), \quad K_{2,3} = K_{5,6} = (EI/L)(\Phi_8/\Delta), \quad (76, 77)$$

$$K_{2,5} = (EI/L)(\Phi_9/\Delta), \quad K_{2,6} = K_{3,5} = (EI/L)(\Phi_{10}/\Delta), \quad (78, 79)$$

$$K_{3,3} = K_{6,6} = (GJ/L)(\Phi_{11}/\Delta), \quad K_{3,6} = (GJ/L)(\Phi_{12}/\Delta), \quad (80, 81)$$

Thus equations (70)–(81), together with equations (42)–(69), give all the terms of the dynamic stiffness matrix  $\mathbf{K}$  of equations (39)–(41). When computing the dynamic stiffness matrix, the terms corresponding to the effects of shear deformation and/or rotatory inertia, i.e.,  $s^2$  and  $r^2$  respectively, (see equations (12)), can optionally be made zero. Thus, for the degenerate case in which  $r^2 = s^2 = 0$ , the computed dynamic stiffness matrix elements of equations (70)–(81) give the same results as those of reference [16].

### 3. APPLICATION OF THE DYNAMIC STIFFNESS MATRIX

#### 3.1. CALCULATION OF NATURAL FREQUENCIES

The dynamic stiffness matrix can be used to compute coupled bending–torsional natural frequencies and mode shapes of composite Timoshenko beams, or of simple structures constructed from such beams; e.g., a non-uniform composite wing. Natural frequencies can be calculated by applying the well-known algorithm of Wittrick and Williams [21], the main features of which are usually used when applying the dynamic stiffness matrix method or its buckling equivalent; see, e.g., the paper by Williams and Wittrick [22]. Basically the algorithm, which is very simple to apply, needs the dynamic stiffness matrices of individual members of a structure and information about their clamped–clamped (i.e., built-in at both ends) natural frequencies. This information is needed to enable the algorithm to guarantee that no natural frequencies of the structure are missed. Thus an explicit expression from which the clamped–clamped natural frequencies can be found facilitates an easy and straightforward application of the algorithm.  $\Delta$  in equation (69) is such an expression, because the clamped–clamped natural frequencies are given by its zeros. Better still, it can be used to give the necessary information about these clamped–clamped natural frequencies without actually finding them; e.g., by isolating them [23]. The Wittrick–Williams algorithm [21, 22] essentially gives the number of natural frequencies of a structure that exist below an arbitrarily chosen trial frequency rather than calculating the natural frequencies. This enables calculation of any natural frequency of the structure to any desirable accuracy.



TABLE 1

*Numerical values of the dynamic stiffness matrix elements of the composite Timoshenko beam of reference [2]*

| Independent stiffness terms | Numerical values at 575Hz |                                |                                           |
|-----------------------------|---------------------------|--------------------------------|-------------------------------------------|
|                             | $r = 0$<br>$s = 0$        | $r = 0.00481882902$<br>$s = 0$ | $r = 0.00481882902$<br>$s = 0.0352784672$ |
| $K_{1,1}$                   | 45945.5751                | 45332.7153                     | 17460.3261                                |
| $K_{1,2}$                   | 1475.17004                | 1461.41919                     | 773.367736                                |
| $K_{1,3}$                   | -3.16770573               | -3.19206133                    | -3.51596822                               |
| $K_{1,4}$                   | -64857.1720               | -64269.5551                    | -37644.4581                               |
| $K_{1,5}$                   | 1529.36699                | 1516.08057                     | 847.825854                                |
| $K_{1,6}$                   | 1.51475697                | 1.48806850                     | 1.12489762                                |
| $K_{2,2}$                   | 44.1874844                | 43.8727065                     | 26.6788227                                |
| $K_{2,3}$                   | 0.210138264               | 0.209582139                    | 0.207397343                               |
| $K_{2,5}$                   | 35.5672039                | 35.2673280                     | 18.5984012                                |
| $K_{2,6}$                   | -0.789925001              | -0.790523037                   | -0.799701447                              |
| $K_{3,3}$                   | 0.246188126               | 0.246186926                    | 0.245714708                               |
| $K_{3,6}$                   | -1.40660574               | -1.40660673                    | -1.40687903                               |

### 3.2. COMMENTS ON THE LIMITATIONS OF THE THEORY

The composite beam considered in this paper is assumed to exhibit coupling between the bending and torsional deformations only. These deformations are assumed not to be coupled with other types of deformation; e.g., extension. Also, the cross-section of the beam is not allowed to warp. The theory assumes that the rigidity properties of the beam are either known or can be found experimentally. These assumptions are quite legitimate for many practical composite beams. Thus although they are severe for open-section composite beams they are much less so for either solid cross-sections such as laminated flat beams with symmetric but unbalanced lay-ups, or for closed-section composite beams such as box beams or aerofoils, for which suitable design results in the only form of coupling present being that between bending and torsion. In addition, the rigidity and other cross-sectional properties for solid or closed section composite beams can be established quite accurately by both theoretical and experimental means [8, 11], whereas open-section composite beams [24, 25] involve much more difficulty. Hence, care should be exercised when using the present theory, particularly when terms related to additional couplings and warping stiffness are significant.

### 4. RESULTS

Numerical results were obtained for the flat glass-epoxy composite beam of rectangular cross-section which formed the illustrative example of reference [2], with all fibre angles set to  $+15^\circ$ . Thus, for computational purposes, the beam is equivalent to a single thick ply. The beam has length = 190.5 mm, width = 12.7 mm and thickness = 3.18 mm. Although reference [2] gives the basic material properties of the beam material, it gives no details of the rigidity properties of the beam or of how they were calculated. Therefore, the authors calculated them independently, assisted by reference [17], as (i) bending rigidity ( $EI$ ) = 0.2865 N m<sup>2</sup>, (ii) torsional rigidity ( $GJ$ ) = 0.1891 N m<sup>2</sup>, (iii) bending-torsion coupling rigidity ( $K$ ) = 0.1143 N m<sup>2</sup>, (iv) mass per unit length ( $m$ ) = 0.0544 kg/m, (v) mass moment of inertia per unit length ( $I_x$ ) =  $0.7770 \times 10^{-6}$  kg m, and (vi) shear rigidity ( $kAG$ ) = 6343.3 N.

TABLE 2  
*C.P.U. time on a VAX 6420 computer using Fortran*

| Number of iterations<br>(number of frequencies) | C.P.U. time (s)      |                  |
|-------------------------------------------------|----------------------|------------------|
|                                                 | Explicit expressions | Inversion method |
| 500                                             | 0.022                | 0.418            |
| 1000                                            | 0.044                | 0.782            |
| 2000                                            | 0.084                | 1.512            |
| 5000                                            | 0.210                | 3.912            |

First, the elements of the dynamic stiffness matrix were computed numerically for several frequencies, by performing the matrix inversion and matrix multiplication steps of equation (41). These were found to agree to machine accuracy with those given by the explicit expressions of equations (70)–(81). Values of the 12 independent stiffnesses are shown in Table 1 for a representative value of the frequency so as to form a comparative datum for readers who wish to check their own composite beam stiffnesses or their computer coding of equations (42)–(81). (For this reason exactly the values of the previous paragraph were used in the data for the computer program; e.g.,  $EI = 0.2865$  means  $EI = 0.28650000\dots$ , etc.) The stiffnesses shown in column 2 of the table agree completely with those given by the authors' earlier theory [16].

Programming the explicit stiffness expressions has a substantial advantage over the numerical method: i.e., the matrix inversion and multiplication steps of equation (41). This is clearly evident from the recorded elapsed c.p.u. times on a VAX computer shown in Table 2. In order to ensure that the times were large enough to be measured accurately, the comparison was made for numerous iterations, each performed at a different frequency. It can be seen that programming the explicit expressions has a more than 17-fold advantage over the inversion method.

Next the first five natural frequencies of the beam with cantilever end-conditions were computed with and without the effects of shear deformation and/or rotatory inertia included (see Table 3), it being noted that these effects can be uniquely described by the parameters  $r$  and  $s$  or  $r^2$  and  $s^2$ . Shear deformation and rotatory inertia are seen to have relatively marginal effects on the natural frequencies of this particular beam. Indeed, the fourth natural frequency of the beam is virtually unaltered due to this frequency corresponding to a torsional mode, for which shear deformation and rotatory inertia would not be expected to have any major effect. However, for certain types of composite

TABLE 3  
*Natural frequencies of the cantilever composite Timoshenko beam of reference [2]*

| Frequency number | Natural frequency (Hz) |                           |                          |                                 |
|------------------|------------------------|---------------------------|--------------------------|---------------------------------|
|                  | $r = 0$<br>$s = 0$     | $r = 0.004819$<br>$s = 0$ | $r = 0$<br>$s = 0.03528$ | $r = 0.004819$<br>$s = 0.03528$ |
| 1                | 30.82                  | 30.81                     | 30.75                    | 30.75                           |
| 2                | 192.7                  | 192.6                     | 189.9                    | 189.8                           |
| 3                | 537.4                  | 536.9                     | 519.2                    | 518.8                           |
| 4                | 648.7                  | 648.7                     | 648.3                    | 648.3                           |
| 5                | 1050                   | 1048                      | 987.5                    | 986.1                           |

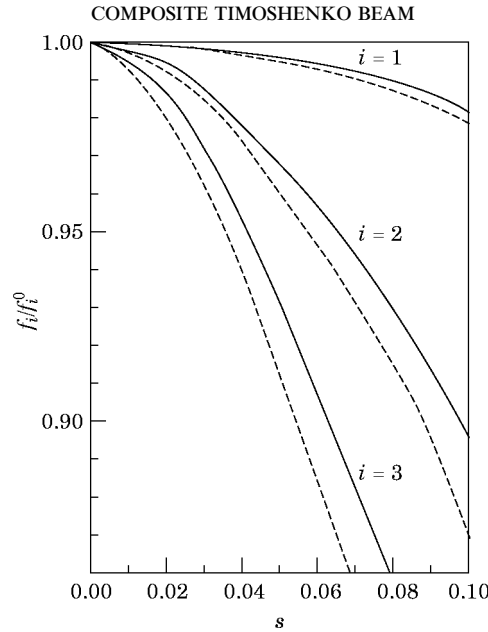


Figure 4. The effects of shear deformation on the natural frequencies of the cantilever composite beam of reference [2]. —, Present theory for  $r=0.004819$ ; ---, curves given in reference [2] for  $r=0.005$ .

beams, the effects of shear deformation and/or rotatory inertia can be much more pronounced. Figures 2–7 of reference [13] illustrate this, but were not used as examples to check the present paper because reference [13] does not account for coupling between the bending and torsional deformations.

The authors of reference [2] have not given the results of Table 3, but in their Figure 2 they have produced a graph showing the effects of shear deformation on the non-dimensional natural frequencies of the beam. Therefore, to validate the current theory, the present authors reconstructed the results of reference [2] using their own data for the beam rigidities. In Figure 4 is shown the effect of shear deformation on the first three non-dimensional natural frequencies  $f_i/f_i^0$  of the beam, where  $f_i$  includes the  $r$  and  $s$  effects, and  $f_i^0$  is the  $i$ th natural frequency which excludes them both: i.e., when  $r = s = 0$ . As expected, the effect of shear deformation is more pronounced for the third mode than for the first two. The agreement between the curves for the present theory and those of reference [2] is acceptable, and the difference is probably due in part to the beam parameters calculated by the authors almost certainly not being identical to those used in reference [2].

## 5. CONCLUSIONS

Explicit expressions for the exact dynamic stiffness matrix elements of a uniform bending–torsion coupled Timoshenko composite beam have been derived; i.e., the effects of shear deformation and rotatory inertia are included. The symbolic computing package *REDUCE* was used in deriving and simplifying these explicit expressions. A related expression for calculating (or alternatively isolating) the clamped–clamped natural frequencies of the beam has also been given to enable an established algorithm to be used to guarantee convergence on all required natural frequencies of free vibration of structures consisting of such beams. The results obtained from the present theory were found to be in good agreement with published results.

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